Program: **Diploma in Artificial Intelligence & Machine Learning**	
Course Code: **6341A**	Course Title: **Deep Learning**
Semester: **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To Provide general introduction to the field of deep neural networks and their applications
- To understand various deep learning methods

Course Prerequisites:

<table>
<tr><th>Topic</th><th>Course code</th><th>Course name</th><th>Semester</th></tr>
<tr><td>Concept of Artificial Intelligence</td><td>3344</td><td>Artificial Intelligence</td><td>3</td></tr>
<tr><td rowspan="2">Concept of Neural Networks</td><td>4342</td><td>Machine Learning & Neural Network</td><td>4</td></tr>
<tr><td></td><td>Soft Computing</td><td>5</td></tr>
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Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarise the fundamental concepts of ANN	14	Understanding
CO2	Outline the concepts of Deep Learning	16	
CO3	Describe Convolutional Neural Networks and Recurrent Neural Networks	16	
CO4	Summarise the applications of deep learning	12	
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped , 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level

CO1	**Summarise the fundamental concepts of ANN**		
M1.01	Recall the basic concepts of ANN	2	Remembering
M1.02	Summarize Activation Functions and Loss Functions.	3	Understanding
M1.03	Demonstrate the training of MLP	2	Understanding
M1.04	Explain Overfitting and underfitting.	2	Understanding
M1.05	Describe Hyperparameters and Validation sets	3	Understanding
M1.06	Explain Stochastic Gradient Descent	2	Understanding

Contents:

Artificial Neural Network: Neural network representation-Structure, Single layer Perceptron, Multilayer Perceptron (MLP), Activation function-Sigmoid, Tanh, ReLU, Softmax. Loss function,
Training MLPs with backpropagation.

Overfitting and underfitting, Hyperparameters and Validation sets, Estimators, Bias and variance. Stochastic Gradient Descent.

CO2	**Outline the concepts of Deep Learning**		
M2.01	Familiarize deep learning	2	Understanding
M2.02	Describe Deep Neural Network algorithm	2	Understanding
M2.03	Initialize network parameters	3	Understanding
M2.04	Explain optimization technique	3	Understanding
M2.05	Explain Regularization Techniques.	3	Understanding
M2.06	Outline Dataset augmentation.	3	Understanding

Contents:
Introduction to deep learning, Deep feed-forward network, Training deep models - Deep Neural Network algorithm, Initialization of network parameters Optimization techniques - Gradient Descent (GD) , Regularization Techniques - L1 and L2 regularization, Early stopping, Dataset augmentation

CO3	**Describe Convolutional Neural Networks and Recurrent Neural Networks**

M3.01	Familiarize CNN	2	Understanding
M3.02	Describe convolution operation.	3	Understanding
M3.03	Familiarize RNN and its types	3	Understanding
M3.04	Demonstrate the steps for training through RNN	3	Understanding
M3.05	Differentiate Deep RNN and Recursive neural networks	3	Understanding
M3.06	Outline the variants of RNN	2	Understanding

Contents:

Convolutional Neural Networks (CNN)- Introduction, Advantages, Convolution operation, Convolutional layers, pooling layers, and padding, Transfer learning and fine-tuning.

Recurrent neural networks – Introduction, Recurrent neuron, RNN unfolding, Types of RNN, Steps for training through RNN, Deep RNN, Recursive neural networks, Variants of RNN-LSTM and GRU.

CO4	**Summarise advanced deep learning architectures**		
M4.01	Demonstrate different applications of deep learning	6	Understanding
M4.02	Demonstrate different applications of deep learning	6	Understanding

Contents:

Advanced Deep Learning Architectures – Autoencoders, different autoencoder architectures, Generative models, Generative Adversarial Networks (GAN), Reinforcement learning

Applications: Images segmentation – Object Detection – Automatic Image Captioning – Image generation with Generative Adversarial Networks (GAN)– Video to Text with LSTM models – Attention models for Computer Vision.

Text / Reference

T/R	Book Title/Author
T1	Ian Goodfellow, YoshuaBengio, Aaron Courville, 'Deep Learning', MIT Press, 2016.
R1	Stone, James, Artificial Intelligence Engines, A tutorial introduction to the mathematics of Deep Learning, Sebtel Press, United State, 2019
R2	Josh Patterson & Adam Gibson, "Deep Learning: A Practitioners Approach", published by O'Reilly Media
R3	Vance, William, Data Science; A Comprehensive Beginners guide to learn the Realms of data science.
R4	Wani M. A, Raj B, Luo F, Dou D, Deep Learning applications Volume3, Springer Publications 2022.

Online Resources

Sl.No	Website Link
1	https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/
2	https://www.ibm.com/deep-learning